

THE EVERCHANGING COMPUTERS

The Evolution of Computers

A **computer** is a programmable computing device that can process, store, and retrieve data. The computer data is processed according to the software's instructions or commands. Computers play an integral part in modern society by transforming the modern world into a global village.

History of Computer

The term “*computer*” originates as a word referring to someone who manually carried out calculations instead of an actual machine.

The 1940s and 1950s

Large machines consisting of thousands of vacuum tubes were part of the early digital computers built in this period. These typically filled an entire room, but the computational power was only a fraction of the functions of personal computers today.

The two distinct families of computing devices are **digital computers** and historical **analog computers**.

- **Digital Computers** – a sequential device that operates data one at a time. The data are represented in the binary format of “0” and “1”; a single transistor is used to symbolize a binary digit.
- **Analog Computers:** The representation of data reflects the properties of the data being modeled in an analog computer. For example, data and numbers may be represented by physical quantities such as electric voltage, whereas digital computers use a stream of binary digits to represent them.

Analog Computers

James Thomson

He did the early foundational work on analog computation in the 19th century. He invented a wheel and a disc integrator used in mechanical analog devices. He then worked with his brother Lord Kelvin to create a device that integrates a product of two functions.

Kelvin then detailed a general-purpose analog machine for integrating linear differential equations of any sequence. He also built a tide-predicting analog computer used at the Port of Liverpool until the 1960s. Tide-predicting machines are used to forecast tide patterns and their unsteady height variations.

Vannevar Bush

He developed the first large-scale, general-purpose analog computer called the “*Differential Analyzer*.” This machine took up a large table and weighed 100 tons, allowing integration and differential equation problems to be quickly resolved.

The *Differential Analyzer* also contained wheels, discs, shafts, and gears to execute calculations and required a considerable set-up time by technicians to solve equations. It contained up to 150 motors and lengthy wires connecting relays and vacuum tubes.

Although the data representation in an analog computer is compact, it is constantly subjected to noise corruption. A single capacitor can symbolize one continuous variable in an analog computer, whereas various transistors are needed in a digital computer.

Analog computers were phased out shortly after World War II.

Digital Computers

Vacuum Tubes (First Generation)

These devices depended on the flow of electric current through a vacuum. Vacuum tubes were widely used in electronic devices such as televisions, radios, and computers until the invention of transistors.

The first generation of computers used thousands of bulky vacuum tubes, with several racks taking up an entire room. The vacuum tube used in the early computers was a three-terminal device consisting of a cathode, grid, and plate.

The first generation of digital computers used vacuum tubes. Examples include the *Atanasoff-Berry Computer* (ABC) developed in 1942, *Colossus* in 1944, *Electronic Numerical Integrator and Computer* (ENIAC) in the mid-1940s, *Universal Automatic Computer I* (UNIVAC I) in 1951; *Whirlwind* in 1951; and the *IBM 701* in 1953.

Transistors (Second Generation)

The transistor is the fundamental building block in modern electronic systems for revolutionizing the field of electronics. Compared to vacuum tubes, transistors were much smaller, cheaper, and more reliable based on functionality.

A transistor is a three-terminal, solid-state electronic device that can control electric current or voltage between two terminals. It is achieved by applying an electric current or voltage to the third terminal. A three-terminal transistor allows an electric switch to be created, which can be controlled by another electrical switch.

The second generation of computers used transistors instead of vacuum tubes, most evidently by the experiments by the University of Manchester. The result of their experiment on the transistor computer was one of the earliest transistor computers whose prototype machine appeared in 1953, with the full-size version commissioned in 1955.

Integrated Circuits (Third Generation)

Jack Kilby of Texas Instruments invented the integrated circuit in 1958. These consists of a set of electronic circuits on a small chip of semiconductor material smaller than a circuit made of independent components.

His idea to properly engineer a piece of germanium to act as several components simultaneously led to the birth of the first integrated circuit. Its progress involved resizing transistors and arranging them on semiconductors. The use of semiconductors led to third-generation computers, with significant advances in speed and efficiency.

Users interacted with third-generation computers using keyboards and monitors and interfaced with an operating system. It allowed the device to process several different applications simultaneously with a central program that monitored the memory. Computers became accessible to a broader audience as they became smaller and cheaper.

Microprocessors (Fourth Generation)

The Intel P4004 1969 microprocessor was the world's first microprocessor. It was the first semiconductor device that provided the functions of a computer at a chip level. It provided the basic building blocks for today's microcomputers, including the arithmetic-logic unit (ALU) and the control unit.

The rise of microprocessors led to the fourth generation of computers with thousands of integrated circuits placed onto a single silicon chip. A single chip could contain components of a computer from the control processing unit (CPU) and memory to input and output controls. Microprocessors could fit in the palm, whereas first-generation computers filled an entire room.

All computers consist of two basic parts: **hardware** and **software**. The hardware is the physical part of the machine, while the software is a set of instructions that commands the computer what to do.

Key Components of Computers

A computer system is not just a single device operating independently. It consists of various components with specific functions and features to support the entire system. These components are divided into three key components: **Input Unit**, **Output Unit**, and **Central Processing Unit (CPU)** – which is further divided into the **memory unit**, **control unit**, and **arithmetic-logic unit (ALU)**.

Input Unit

A computer system responds to the instructions it receives from the users and uses an input unit to acquire those instructions. Input units take user commands in the form of data and convert them into a digital language format sent to the processing unit for the next step.

Examples of input units include a keyboard, mouse, scanner, joystick, and trackball.

Central Processing Unit (CPU)

The data from the input unit proceeds to the processing unit before it is executed. CPU understands the command and how it can be achieved. It converts the data accepted in human language into machine language for computers to interpret.

The CPU has three (3) main components:

- a. **Memory Unit:** When data reaches the processor from input devices, the memory unit saves it immediately. It has pre-existing programs that transmit the data to the other parts of the CPU.

It is where all the information becomes available, and bits and bytes (8 bits) are used to measure data size.

This unit is also responsible for storing data from input devices and storing the result data from output devices.

- b. **Control Unit:** This manages the computer device functions and is the central component of the CPU.

Once the data is in the memory, it processes it for execution. It is where the data conversion from human language to machine language occurs as it interprets the signal and sends it over to the output unit.

This unit maintains a data flow inside the system and transfers the commands to the arithmetic-logic unit.

- c. **Arithmetic-Logical Unit (ALU):** This processor is responsible for mathematical calculations and issues the computer system deals with while functioning. It is also used for data comparison and decision-making.

It has features to accomplish different mathematical solutions like addition, subtraction, and multiplication, but it only performs the calculation when necessary and then sends it to the output unit.

Output Unit

When the input and central processing unit processes are accomplished, the data is fed to the output unit, where the result is shown. These devices always connect to the system, thus making coordination easier.

A computer monitor is one of the primary output devices that displays the results to the user. Other examples include printers, projectors, speakers, headphones, and plotters.

The Operating System

Operating System (OS) – a collection of software programs and commands that control computer hardware to make it usable. It allows the computing power of the hardware available to the computer users and manages the hardware to attain good system performance.

An operating system handles hardware such as the processors, input/output devices, communication devices, and data, and it provides functionality, including hardware sharing, error recovery, and network communication management.

Advantages of Operating System

- **Abstraction** – allow users to hide details of hardware by creating an abstraction.
- **Executable Programs** offer an environment where users may execute programs or applications.
- **User-friendly** – provides the computer system with easy-to-use format resources.

Disadvantages of Operating System

- **Volatility**: Users may lose all the contents stored in the computer system whenever an issue occurs.
- **Expensive**: OS software is costly for small organizations, which can cause an additional financial burden.
- **Unpredictable**: It is never secure, as threats can happen anytime.

Examples of Operating Systems

Microsoft Windows

It is a family of graphical operating systems created by Microsoft. The original Windows 1.0 OS was introduced in 1985 as a graphical operating system shell to compete with Apple's GUI OS in 1984.

Currently, Microsoft Windows wins the personal computer and laptop market with over 90% market share. However, Windows has not been as successful in dominating the smartphone and tablet market.

Mac OS

It is the desktop operating system created by Apple Inc., designed to work for Macintosh computers. Mac OS provides several different features, such as a graphical user interface, multitasking, and memory security.

Mobile Operating Systems

Android

It is a mobile operating system developed by Google and the Open Handset Alliance designed for touchscreen smartphones and tablets. The first Android smartphone was released in late 2008, and Android remains the most widely used mobile operating system.

iOS

It is a mobile operating system for Apple's smartphones and tablets. It was developed from the MAC OS/X operating system in 2007. Multitasking for iOS was introduced in 2010 with the release of iOS version 4.0.

References:

Data Flair (n.d.a.). Components of a computer and their functions.
[Web Article]. Retrieved on July 5, 2022, from <https://data-flair.training/blogs/computer-components-with-functions/>

O'Regan, G. (2021). A brief history of computing. Springer.

The Avinash Pandey (2022). 10 examples of operating systems.
[Web Article]. Retrieved on July 5, 2022, from
<https://quicklearncomputer.com/examples-of-operating-system-types-of-operating-system>